

No. of Functional Features Included in the Solution

Date	11 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Allows users to register, log in, and securely access their accounts using JWT and Firebase Authentication.	Authentication	Done	10
2	Study Material Upload	Enables users to upload PDF, DOCX, TXT files or paste raw text for AI processing.	Materials	Done	10
3	AI Summary Generator	Automatically generates concise and structured summaries from uploaded study materials.	Summaries	Done	15
4	AI Flashcard Generator	Creates interactive flashcards for efficient revision and self-learning.	Flashcards	Done	10
5	AI Quiz Generator	Generates MCQ, True/False, and Short Answer quizzes with automatic evaluation.	Quizzes	Done	15
6	Personalized Study Planner	Creates customized 7-	Study Plans	Done	10

		day and 30-day study plans based on learning goals and performance.			
7	Learning Analytics Dashboard	Displays quiz scores, progress charts, study activity, and weak topic analysis.	Analytics	Done	15
8	Firebase & Local Storage Integration	Stores user data securely in Firebase Firestore with Local JSON fallback for development.	Database & Storage	Done	15

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Authentication, Material Upload, AI Summary, AI Flashcards, AI Quiz Generator, Study Planner)
Additional / Nice-to-Have Features	2 (Learning Analytics Dashboard, Firebase with Local JSON Fallback)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Responsive user interface, dashboard, navigation, file upload, theme support	Login/Register, Dashboard, Material Upload, AI Summary Page, Flashcards, Quiz Page, Study Planner, Analytics Dashboard
2	Backend / Logic	Business logic and AI processing	Authentication, File Processing, Summary Generation, Flashcard Generation, Quiz Generation, Study Plan Generation, Analytics Engine
3	Database / Storage	Data management and storage	Firebase Firestore, Local JSON Fallback, User Data, Study Materials, Quiz Results, Analytics Data
4	API / Integration	External services and API communication	Groq API (Llama 3.3-70B), Firebase

			Authentication, Firebase Firestore, REST APIs, Axios Integration
5	Security / Authentication	User authentication and application security	JWT Authentication, Firebase Authentication, Password Validation, Protected Routes, CORS Configuration